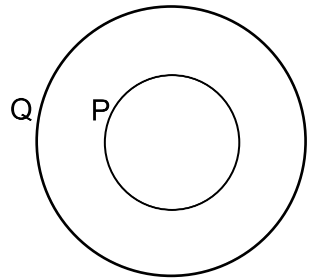
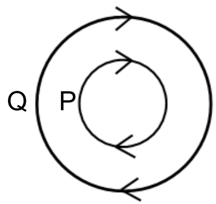
23 Two circular coils P and Q, each of a single turn, have radii 0.10 m and 0.20 m respectively. They are arranged concentrically as shown. Coil P carries a current of 3.0 A, while coil Q carries 5.0 A. The magnetic flux density at the center of a single-turn circular coil is given by $B = \frac{\mu_o I}{2r}$, where I is the current in the coil of radius r and μ_o is the permeability of free space.

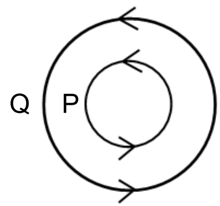


If the resultant magnetic flux density at the centre of the coils is $2.5\,\mu_o$, directed out of the page, which diagram shows the correct directions of the currents in the coils?

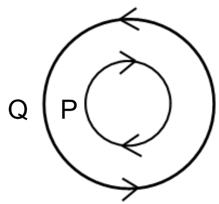
A



B



C



D